Strategic Decision Making Support Model on RTE Approach from the BPM

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ABSTRACT
RTE is a new management paradigm on which core processes are executed and managed with the latest information. This concept is a needful concept to continuously strengthen corporate competitiveness in an era of unbridled competition with the collapse of traditional value chain and accelerated flow of information. Paradigm of AI (Artificial Intelligence), which is based on the assumption that existing business processing types or expertise in a predefined and limited management environment can help to make a right decision in the new future environment, and expert system should overcome rule based and model based limitations. In RTE, rather than existing business types or expertise, it is important to monitor the event or information at the present point of time that can provide support in the current situation.

In this research, a model that integrates KM (Knowledge Management), BI (Business Intelligence) with business process is presented, which are essential for decision making in a real-time based corporate environment. Furthermore, in a dynamic management environment of real world, a decision making support model that can overcome three limitations that are posed by AI, Expert system paradigm shall be presented in this research.

Keywords
RTE; Knowledge Management; Decision Making; System Dynamics; Business Process Management

1. INTRODUCTION
With accelerated liberalization and globalization of management environment and rapid advancement of IT, the changing speed of corporate environment is getting faster. As a result, corporate environment is veering to challenging environment. The structural characteristics of the 21st market in such environment can be summarized into the followings; 1) rapid change, 2) high uncertainty. In this corporate environment, RTE (Real Time Enterprise) is drawing attention. According to McGee of Gartner (2001), who advocated the concept of RTE, RTE is "a management system that maximizes competitive edge by continuously removing delay factors in managing and executing core business processes, based on the latest information." If a company can identify, analyze an internal or external change, share such information, make a right decision, and take subsequent measures on a real time based or within an acceptable short period of time in dynamically changing corporate environment with uncertainty, it is RTE.

ICEC'05, August 15–17, 2005, Xi'an, China.

In the meantime, many companies have concentrated on the development of intelligent management system that can lay a perpetual foundation for competitive edge in the future as they have been faced with uncertain and unforeseeable business environment [28]. Knowledge management can be defined as the one that identifies various level of knowledge of individual, team, organization [35,7]. Many of KM methodologies that have been used emphasized improvement of business process [7,15,23] and acquisition of competitive benefit [35,47], concentrating on the structure to effectively and comprehensively implement KM [49,13,27]. Information processing in existing KM has been focused on knowledge management system based on AI (Artificial Intelligence) and Expert system [28].

Decision making process in an organization is supported by many knowledge bases generated from various experience and perspectives [42]. Actual task in knowledge management is to link and integrate knowledge to be applied with the internal and external environment of a company, and then analyze and extract knowledge [6]. Knowledge management is an important factor for RTE, and at several phases, it has a close relationship with RTE. At the RTE implementation phase, quality of RTE is greatly affected by the quality of BPM (Business Process Management) and Explicit Knowledge that supports business cooperation. BPM is one of the means to make an organization get an insight into knowledge based perspective [15]. From the point of time when RTE is in operation, the depth of human collaboration between employees & companies and Implicit Knowledge decide the success of management. Therefore, an organization is needed to be sensitive to such knowledge, flexibly and rapidly changing environment.

In a predefined and limited corporate environment, the paradigm of AI (Artificial Intelligence) and Expert system, which are based on the assumption that existing business processing or expertise help to make a right decision in a new future environment, should overcome the rule based and model based limitations [28]. Malhotra (2001) presented 3 areas that should be overcome; *first*, according to the concept of AI and Expert System, AI and Expert System based knowledge management techniques can deliver right information to a right person at a right time. But, in RTE environment, the meaning of 'right' changes as time goes by and as environment changes; *second*, AI and Expert system technologies are based on human intelligence and experience. Existing database and groupware application store static information and documents. However, such information and document cannot store their rich schema that has been imbedded in the course of generating such information and document. Such static expression of information cannot dynamically provide linkage with dynamic corporate environment and increasing number of events. Besides, even if same information is extracted, interpretation on the information might differ according to a person who extracts it and according to time; thirdly, it is said that AI and Expert System technologies can accurately forecast information to be released and who need this information. However, how many irrelevant information or mails do we receive in our mail box or personal portal screen!

In this study, decision making support model that can overcome the 3 aforementioned limitations in actual dynamic management environment shall be presented and how to implement such model shall be proposed. Even though many themes related to BPM from strategic perspective [2] can be presented, this study shall focus on the linkage with knowledge management.

2. LITERATURE REVIEW

2.1 Knowledge based decision making in RTE

The researches have been made on traditional decision making [39], behavioral decision making [11,21], judgment and decision making [30], systematic decision making [29], and recently NDM (naturalist decision making) [24,51]. The development of the aforementioned studies shows that static researches in laboratories have moved toward the researches on dynamic and business areas. Whether decision is made on general businesses or whether strategic decision is made resulted in different methodologies. Metrics to be applied to different decision making environment, various perspectives, knowledge mode have been developed [Figure 1].

In static and experimental decision making environment, AI (Artificial Intelligence) approach [31,32,33], and decision making tree & cognitive mapping [4] have been mainly used for traditional decision making.

Knowledge based Expert system showed its effectiveness for operation and explicit decision making, but, has restrictions in strategic decision making [10,28]. Expert System as an adjunctive system like advisory system can help to make a better decision, but final decision is made by decision maker. AI focuses on well defined problems, but is confined to a certain scope of problems that can take place in the limited scope of relationship with objectives and characteristics. Even though Expert Systems have been developed or strategic planning, most of them cannot model the structure of strategic problems. Expert system defines the structure of expert knowledge by integrating traditional computer system, in particular, DB and spreadsheet based system, financial system, modeling, forecasting and reporting system [26,32].

In cognitive modeling, cognitive map is one of the widely used techniques. This approach is expressed in diverse concepts--cognitive map [4,25,48]; casual map [9,16]; influence diagram [8,38]; knowledge map [18]. Cognitive map shows the cognitive relationship between concepts of the given environment and attributes. This model does not consider the factor of time, while cause and effect concept has an impact on exceeding times. Therefore, cognitive map has a restriction since it cannot explain dynamic characteristics of real world.

In NDM (Naturalistic Decision Making) environment, most of general methodologies for dynamic and practical commercial decision making environment are CTA (Cognitive Task Analysis) [14,24], RPD (Recognition-primed Decision) model [20, 24], and SA(Situation Awareness) [24]. NDM researchers are trying to understand "knowledge in the real world."

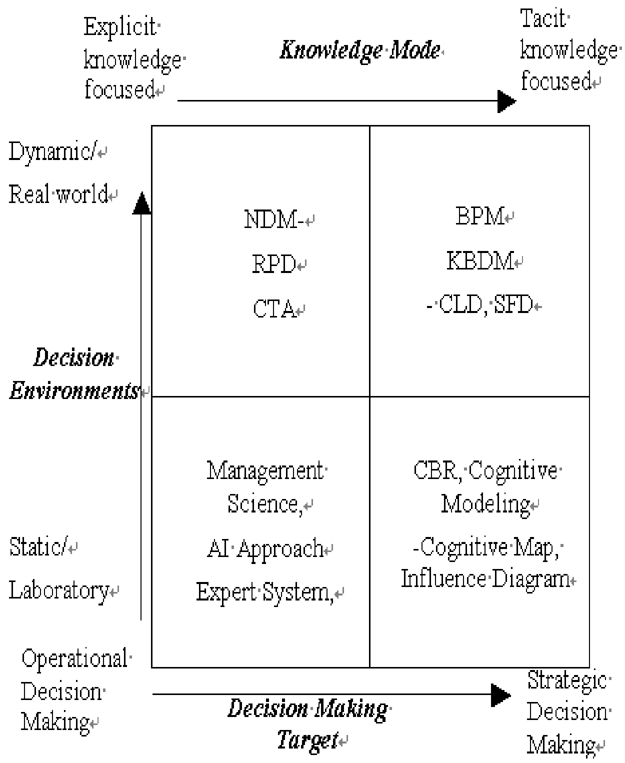

Figure 1 –Categorization of application approach based on decision making environment

However, general focal point of NDM is on professional technology and operational level rather than strategic decision making. Even though there are various causes and reasons behind most of the decision made, most of the decision makers don't understand them or have know idea about them. As a result, they cannot easily define the relationship between the cause and effect. NDM cannot check out cause-effect relationship and feedback of knowledge and impact of time. The task of NDM is to develop simulation methodologies, strengthen the method to extract knowledge for decision making in actual conditions [17]. RPD explains how experience of making difficult decision are used and insists that it can make an effective decision without logically selected strategies [24]. Experts use their experience to form simulation on the recently created problem and to present proper solution [5]. It is similar to CBR (Case-based Reasoning) strategy that uses the theory of Expert System.

In Particular, even though strategic decision making would differ based on propensity of decision makers, the environment of strategic decision making becomes closer to the dynamic real world and implicit knowledge is used [6,10]. Therefore, a model which is an improved knowledge based decision making and its implementation method shall be presented to support decision makers in their decision making or strategic planning by applying BPM in order to overcome such limitations.

2.2 Organizational memory and organizational learning

Organizational memory in an organization, for example, is composed of problem solution & technique, experience of human resource, experience of business processing and know-how of lesson learned. OM (Organizational memory) is what scattered know-how is closely integrated.

OM can be defined with technical perspective, system perspective [44], human resource perspective [40,41], their combined perspective [1]. Schein (1996) defines organizational culture as embedded memory. And Stein (1995) defines OM, saying that historically obtained knowledge is related to the current activities and have an impact on them, and OM is a mean to bring about organizational effectiveness however the impact level is high or low. From a most general perspective, OM is related to reuse of organizational experience [1]. Most of OM definitions make mention of storage layer and facilitation layer. However, the perspectives so far excluded the interactive relationship between information system and people to improve organizational effectiveness. OM is an incidental effect of learning. That is to say, OM is derived from works in community. OM and organizational learning cannot be separated and they are closely related. Acquisition and maintenance, which are the OM processes, focus on knowledge base of organization. Ackerman and Halverson (2000) said that organizational memory can be regarded as a kind of perfect and unique repository of whole organization. We define this as integrated knowledge model in this study.

Most of valuable knowledge is embedded in mental model of individual. The trickiest part in OM is expression of such conceptual model. Conceptual model of individuals contain most of organizational knowledge (know-how and know-why). Consequently, representation of such knowledge becomes the major factor for making a right decision. In previous researches, implicit knowledge was expressed in impact [18], cognitive map [34], ERC [16] and RPD [24]. However, these methods show only concept and figure (map). They have limitations in expressing feedback and operational knowledge. Kim (1993) explained that organizational learning is dependent on improvement of conceptual model of individuals, and externalization of these conceptual models is important to develop new shared conceptual model. But, conceptual model is mixture of what is clearly learned and is allusively merged. Though, his learning model cannot explain dynamic and non-linear phenomenon.

The problem faced by organization is that it is difficult to establish excellent organizational memory of its own since each organization has its unique OM [3]. Atwood(2002) explains 4 major tasks related to organizational memory: maintenance of process perspective, maintenance of context of document, provision of related knowledge, maintenance of social context. In this study, we present knowledge based decision making model from process management perspective as a mean to accomplish such major tasks.

3. KNOWLEDGE BASED DECISION MAKING MODEL IN RTE

In RTE, decision making or business execution is done by the partnership of people, process, and system. Hence, in order to achieve effective management performance in real time corporate environment, cooperative system among customer oriented service provision, BPM (Business Process Management) required to provide such services, knowledge management, BI (Business Intelligence) are required.

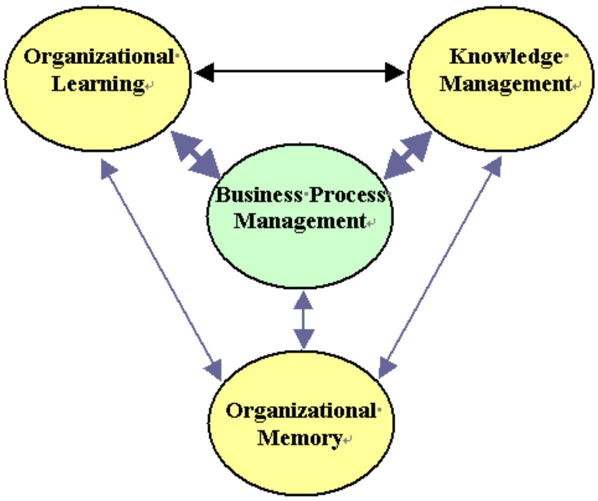

Figure 2 – Link between organizational learning and knowledge management

In order to effectively operate such system, events related business activities should be detected, related feedback should be provided, and workplace should be provided for effective communication and facilitation of feedback. At the same time, in order to smoothly support business activities, BPM to support cooperation among people, process, application, and BI in which required knowledge is managed, newly extracted knowledge is stored are required [Figure 2]. Process management provides frameworks for organizational learning and enables knowledge management [28]. Accordingly, BPM provides a mean to generate and extract knowledge as well as a mean to search knowledge required for business activities.

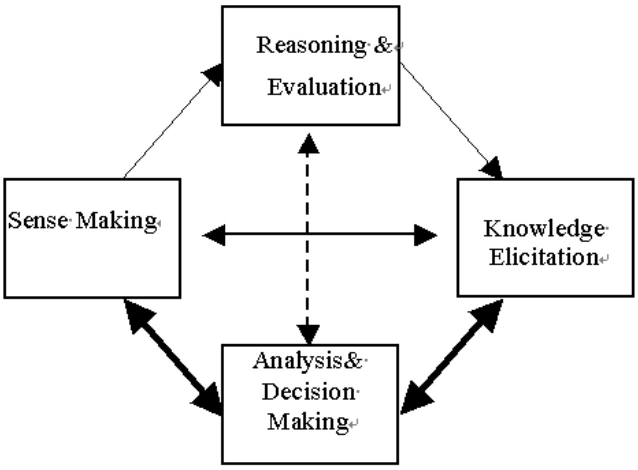

Figure 3 – BPM oriented decision making

In the course of executing such businesses, learned knowledge is managed as individual knowledge, and organizational memory is extracted from individual knowledge and information system. Organizational memory is a mean to conduct current activities through historically obtained knowledge, which impacts on organizational effectiveness [43]. [Figure 3] model is suggested to implement BPM oriented decision making support system. Here in this study, the following methods for analysis and decision making in Figure 3 are suggested.

4. METHOD OF KNOWLEDGE BASED DECISION MAKING IN RTE

In order to overcome rule and model based limitations of AI and Expert System paradigms, a method against dynamic and uncertain decision making environment is required. That is to say, model should be transformed from the knowledge based model on which historical experience or historical data are based to the knowledge based model that sufficiently reflects dynamics, events, which are related to the current decision making to decision making elements. Rather than historical experience, it is more required to have the knowledge base that responds to current event and uncertainty in the future and to establish a system to be able to monitor current events.

4.1 Knowledge based decision making model

Even though in decision making process, application of knowledge [14,16,36] is a very tricky research theme, it is very important and meaningful in actual corporate management. The suggested method of knowledge based decision making in this study supports right decision making in a dynamic and special circumstance. The model that links knowledge management with decision making in knowledge based decision making model is presented in Figure 4.

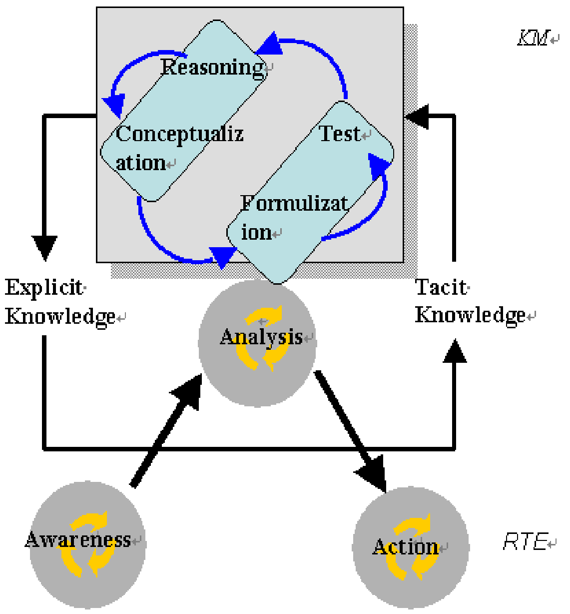

Figure 4 – Knowledge based decision making in RTE

While RTE responds to customer (internal and external customer) and market requirement, continuously monitor them, it detects changes and put them as events. Corporate operation system and 6 sigma activities act as a sensor to detect events at the operational business level. Changes in event at strategic business level show different behavior from changes at operational business level [50]. As a business is processed for the event detected through the aforementioned way and decision is being made, learning cycle and knowledge transfer come into the scene [9]. By using KM, the identified event can be reasonably handled and a reasonable decision can be made in a dynamic environment.

The difference in RTE from existing decision making model is to be able to make decisions against actual events by utilizing related business document, explicit knowledge and tacit knowledge generated in the course of executing business process.

4.2 How to support knowledge based decision making

knowledge based decision making consists of 5 phases; 1) problem recognition and definition, 2) integration with conceptualization of knowledge, 3) formulation of knowledge model, 4) establishment of monitoring system for operational measures, and 5) decision making support.

4.2.1 Phase 1. Problem recognition and definition

Firstly, fundamental problem faced by a company should be recognized. If such problems are identified, a company can set the direction of implementing RTE. Since internal and external environments of all the companies are different, a company should identify its problems first in order to find proper a model that suits its organization.

Secondly, the company should identify where the fundamental problems were created. Management innovation methodologies such as six sigma can be applied to identify such problems and come up with measures against them.

Thirdly, by checking out if the identified problems can be resolved on a real time basis, the company should set the scope of real time implementation.

Fundamental problems are generated due to inefficient access or lack of acquisition of core information. Hence, it is imperative more than anything else to secure accessibility to information. Furthermore, it is important to have a touchless process from the phase of detecting latest information to the information control phase. Acquisition of such essential information plays a role in decreasing incidental corporate capacity and inventory.

4.2.2 Phase 2: Integration with conceptualization of knowledge

Based on identified and defined problems, business processing related to explicit knowledge is extracted, and relationship between dynamics and assumption and their related factors shall be defined. It is needed to extract related knowledge and define the relationship between those factors and management performance measurement metrics.

Integration of business process related explicit knowledge

The role of KM in RTE is to code obtained employees' knowledge and to make it shared. And, the eventual objectives of BPM and sophisticated business application are to automate what people do and duplicate it in a way that others can do. However, since it is impossible to perfectly replicate them, what BPM tries to do is to generate business processing model for business activities and reflect the executed activities identically.

All the matters related to such businesses are coded and expressed in business rules, data elements, metadata structure.

Definition of dynamics relationship

It is essential to determine the relationship in assumptions of dynamics. Assumptions for dynamics can be conceptualized through CLD (Casual Loop Diagram) in order to represent the cause and effect relationship among factors. Based on problems identified in the Phase 1 and situation Dynamics assumptions can be derived from focus group interview, discussion on actual events, and interview with experts. However, most of individuals or functional departments keep limited knowledge. Therefore, it is required to reorganize partial knowledge obtained from different sources, consolidate it into integrated knowledge model to conceptualize management objectives and related problems. At this stage, explicit or implicit presentation of problems is structuralized through CLD and knowledge on procedures of business processing process is defined, using process map.

4.2.3 Phase 3. Formalization of knowledge model

At the formalization phase, based on conceptualized CLD, factors and variables that have an impact on decision making elements are formalized through SFD (Stock & Flow Diagram) [46]. For formalization, it is required to identify the variables related to quantity in order to put those factors and variables in formula. In order to review and finalize the identified variables, simulation is conducted and data is statistically adjusted. To do so, fundamental data such as coefficient, rates should be collected from corporate reports. The knowledge model conceptualized in Phase 2 is sophisticated in this phase. Next, it is necessary to prove the model in order to see if formalized model represents the conceptualized model well and this model shows same trend of actual historical changes. For this verification, diverse methods can be applied; verification on feedback loop by decision maker, comparison between simulation result and historical data, verification on model in an extreme situation, sensitivity analysis against various variables [37,45].

4.2.4 Phase 4. Establishment of monitoring system for operational measures

In this phase, problems identified and defined in Phase 1 and related operational measures are defined and relationship with KPI (Key Performance Indicator) related to operational measures is defined. And, which process is related to the defined operational measures is defined and the relationship between operational measures and business dynamics are defined. Furthermore, management system for them is established. Generally, even though value driver tree is defined from the KPI perspective and drivers that have an impact on corporate values are analyzed starting from financial drivers to operational drivers, it cannot monitor and manage operational measures. In existing KPI operation, cause and effect relationship between higher and lower KPIs are taken into consideration and KPIs is under management by identifying KPI pool.

4.2.5 Phase 5. Decision making support

At RTE management level, the focal point of KM moves toward shared implicit knowledge and business cooperation. Decision in RTE is made based on the events detected through real-time monitoring. Therefore, real time information should be delivered to a right decision maker and at a right time. To do so, the related decision makers and experts should be identified, and their profiles should be under management and mapped with business process.

Organizational learning and organizational memory

These types of decision makings take place in the workplace, e.g. enterprise portal. And the events are to relate to process respectively. However, the perspectives so far excluded the interactive relationship between information system and people to improve organizational effectiveness. OM is derived from works in the community in workplace. We defined this as integrated knowledge model in this study. In this phase, OL and OM are created naturally during the work.

4.3 Difference from existing research

Approaches for existing decision making system have been based on AI. In these approaches, decisions in the current situation are made, based on the pattern or information that seems to be most similar to historical cases or rules. However, the decision making model suggested by this study is different since it considers the current event information related to the task on which decision is to be made as well as historical pattern, and in this model measures can be taken through detecting problems in advance. While existing decision making model has been operator oriented decision making support model, the model in this study can support even decision making of managers and executives.

5. CASE STUDY

In this case study, details related to Phase 4 and Phase 5 of how to establish knowledge based decision making are described. Based on the aforementioned phases, S company established abnormality processing business based on BPM. Staffware iPEV2.7 was used as BPM solution. This company established a process to support BPM oriented decision making system suggested in [Figure 4]. It was implemented in a way to monitor operational measures through BPM from workplace and for decision maker to be able to make a decision on the expected problems when an event takes place. At the same time, the knowledge identified during decision making was stored in knowledge base through KMS. Also, explicit knowledge, which is needed for decision making, was used from existing data or information through BI as well as knowledge base.

As [Figure 6] shows, through establishment of operational measures monitoring system of Phase 4, all the events were identified at actual operation level and such events and information were reflect to decision making.

To establish such monitoring system, business process and operational measures were defined and relevant business rules and business dynamics were defined. Besides, the system provides support in a way that a person who can have the best judgment on the related business was identified and decision maker was able to make a real-time decision.

At the decision making support of Phase 5, as shown in [Figure 6], while process progress was being monitored, business at the bottleneck where problem occurred were handled and

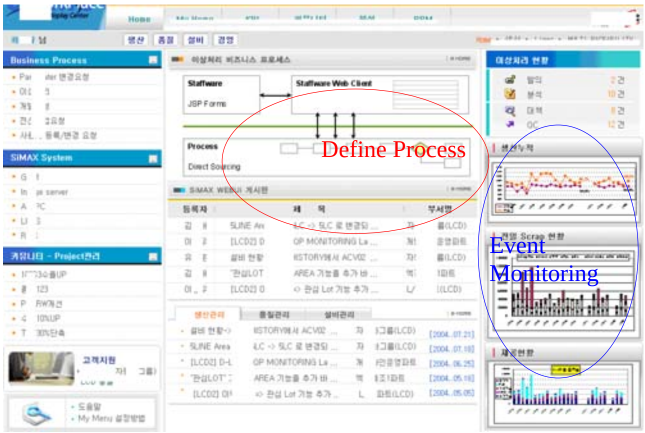

Figure 5 – BPM oriented decision making support system

controlled. At this phase, individual business processing tacit was shared, and messages on issue handling were delivered, they were notified to a right decision maker, linkage with related decision making elements was made in order to resolve the problems on a real time basis. Besides, through post analysis and tracking, related implicit knowledge was derived, failure occurred in the process and the areas to be expected to have problems were able to be forecasted.

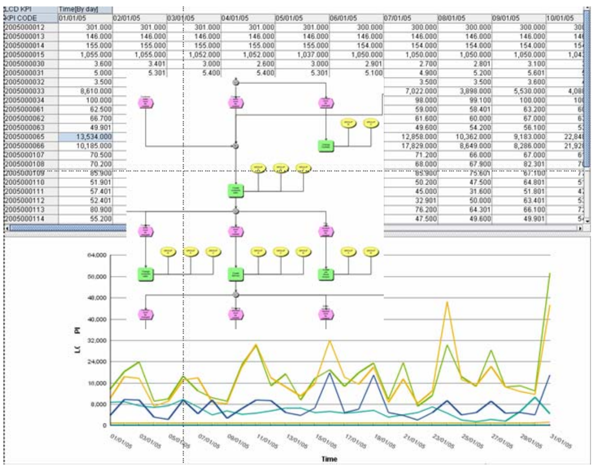

Figure 6 – Operational measure monitor

6. CONCLUSION

For the decision making in dynamic and uncertain decision making environment, it is needed to cast off the existing information processing knowledge management of AI and Expert system paradigms. And if a decision is made through searching the current tasks on which decision is to be decided and related event knowledge on a real-time basis, a right decision for the current problem can be made. That is to say, model should be switched over from the knowledge based model based on past experience or generated historical data to the knowledge based model that reflects dynamics and sufficiently reflect current event, which are related to decision making, to decision making elements.

In order to make effective knowledge based decision in a dynamic and uncertain corporate environment, such changes should be detected and a right decision maker should make a right decision to take measures against those changes at a right time. To do so, dynamics related to business processing as well as process should be under management. Such factors should be reflected to decision making.

This study focused on knowledge based decision making model in RTE environment and its implementation method. The future research is needed on analysis on ROI of "operational measures" in RTE and development of integrated model between business process and dynamics.

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